

Fuselage Mission Planner — Tooling, hosting & repository

Reference for languages, IDE, SDK locations, dependencies, and where code and services run. Sourced from this repository, `git remote`, `pubspec.yaml`, `backend/package.json`, `.github/workflows`, and workspace Cursor rules.

1. Source repository (GitHub)

- Origin (clone URL):**
`https://github.com/fuselagecursor-maker/mission-planner.git`
- Web UI:**
`https://mission-planner-coral.vercel.app/`
- Local workspace (this machine):** `D:\mission planner`

2. Languages & markup

Language / format	Where used
Dart (SDK <code>>=3.4.0 <4.0.0</code> per <code>pubspec.yaml</code>)	Flutter application under <code>lib/</code> , tests under <code>test/</code>
JavaScript (Node, CommonJS)	Backend under <code>backend/src/</code> (<code>server.js</code> , <code>auth.js</code> , <code>db.js</code> , <code>migrate.js</code>)
SQL	PostgreSQL schema in <code>backend/sql/</code> , migrations
YAML	<code>pubspec.yaml</code> , <code>analysis_options.yaml</code> (repo root), GitHub Actions workflows
JSON	<code>backend/package.json</code> , API payloads
Markdown	<code>README.md</code> , <code>docs/*.md</code>
HTML / CSS	Flutter web output (<code>build/web</code> when built for web)

3. IDE & editor tooling

Tool	Role
Cursor	Primary IDE for this workspace; AI-assisted editing; VS Code-compatible extensions and UI.
Workspace rules	<code>.cursor/rules/</code> (e.g. D-drive tooling policy for caches/SDKs).
Flutter / Dart tooling	<code>flutter</code> , <code>dart</code> CLI; <code>flutter pub get</code> , <code>flutter run</code> , <code>flutter build</code> ; Dart analyzer & <code>flutter_lints</code> (dev dependency).
Node tooling	<code>npm install</code> , <code>npm run start</code> in <code>backend/</code> .

Tool	Role
Git	Version control; remote <code>origin</code> → GitHub URL above.

4. Frameworks & runtimes

Component	Technology
Mobile / desktop / web UI	Flutter (Material Design)
HTTP API	Express 5 (<code>backend/package.json</code>)
Database	PostgreSQL (driver: <code>pg</code>)
Auth	JWT (<code>jsonwebtoken</code>), bcryptjs password hashing
CORS / env	<code>cors</code> , <code>dotenv</code>

5. Where SDKs, caches & installs live (this workspace policy)

From `.cursor/rules/d-drive-tooling.mdc` — tooling is configured to prefer **D:** over large installs on **C:**:

Item	Typical path / env
Flutter SDK	<code>D:\flutter</code>
Dart / Pub cache	<code>D:\dev\pub-cache</code> (<code>PUB_CACHE</code>)
Android SDK	<code>D:\dev\android-sdk</code> (<code>ANDROID_HOME</code>)
Android AVDs	<code>D:\dev\android-avd</code> (<code>ANDROID_AVD_HOME</code>)
JDK (Gradle / Android)	<code>D:\dev\jdk-17.0.18.8-hotspot</code> (<code>JAVA_HOME</code>)
Gradle user home	<code>D:\dev\gradle</code> (<code>GRADLE_USER_HOME</code>)
Node global prefix (policy)	<code>D:\dev\npm</code>
pip cache (policy)	<code>D:\dev\pip-cache</code> (<code>PIP_CACHE_DIR</code>)
Temp for installers/CLI (policy)	<code>D:\dev\temp</code> (<code>TEMP</code> / <code>TMP</code>)

Project source: application and backend code live under `D:\mission_planner\` (not an SDK install path).

6. Application & dependency inventory

6.1 Flutter

- **SDK:** Flutter (pinned channel per your local install; project requires Dart `>=3.4.0 <4.0.0`).
- **Dependencies:** `cupertino_icons` , `google_fonts` , `camera` , `flutter_map` , `latlong2` , `shared_preferences` , `geolocator` , `http` .
- **Dev:** `flutter_test` , `flutter_lints` `^4.0.0`.

6.2 Node backend (backend/package.json)

- `express ^5.2.1`, `pg ^8.20.0`, `jsonwebtoken ^9.0.3`, `bcryptjs ^3.0.3`, `cors ^2.8.6`, `dotenv ^17.4.2`.
- **Scripts:** `npm run start` / `npm run dev` → `node src/server.js` .

7. Where services & data run (hosting)

7.1 Local development (defaults in repo)

Artifact	Host / URL	Notes
Flutter app	Device, emulator, or Chrome (web)	<code>flutter run</code> ; web build output in <code>build/web</code>
Auth API	<code>http://localhost:4000</code>	<code>PORT</code> from env or default 4000 (<code>backend/src/server.js</code>)
Flutter → API URL	Default <code>http://localhost:4000</code>	AuthApi uses <code>--dart-define=AUTH_API_URL=...</code> to override
PostgreSQL	Local or remote host from <code>backend/.env</code>	Database name documented as <code>mission_planner</code> in <code>backend/README.md</code>

7.2 CI / optional cloud (documented in repository)

File `.github/workflows/deploy-web-vercel.yml` (manual `workflow_dispatch`):

- **Runner:** `ubuntu-latest` .
- **Steps:** `actions/checkout` , `subosito/flutter-action` (stable), `flutter pub get` , `flutter build web --release` with `AUTH_API_URL` from GitHub Actions secret.
- **Deploy target: Vercel** (requires `VERCEL_TOKEN` secret); comments reference example API on **Railway** and CORS alignment.

Important: Actual production URLs depend on your configured secrets and Vercel project; they are not hard-coded in application source.

8. Document & diagram PDFs (this repo)

- `docs/SRS_Fuselage_Mission_Planner.md` / `.html` / `.pdf` — SRS (Python `markdown` + Edge print).
- `docs/arch/Fuselage_System_Architecture_and_Use_Cases.html` / `.pdf` — architecture + UML figures (`build_architecture_document.py`).
- `docs/arch/Fuselage_Tooling_Hosting_and_Repository.html` / `.pdf` — this document (`build_tooling_hosting_pdf.py`).

9. Quick command reference

Goal	Command / location
Flutter dependencies	<code>flutter pub get</code> (repo root)
Run mobile app	<code>flutter run</code>
Backend dependencies	<code>cd backend && npm install</code>

Goal	Command / location
Run API	<code>cd backend && npm run start</code>
Apply DB schema	Run <code>backend/sql/init.sql</code> on PostgreSQL (see <code>backend/README.md</code>)

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